

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv('mymoviedb.csv', lineterminator = '\n')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   object
1   Title                 9827 non-null   object
2   Overview              9827 non-null   object
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   object
7   Genre                 9827 non-null   object
8   Poster_Url           9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
In [5]: df['Genre'].head()
```

```
Out[5]: 0    Action, Adventure, Science Fiction
        1           Crime, Mystery, Thriller
        2                      Thriller
        3    Animation, Comedy, Family, Fantasy
        4    Action, Adventure, Thriller, War
        Name: Genre, dtype: object
```

```
In [6]: df.duplicated().sum()
```

```
Out[6]: np.int64(0)
```

```
In [7]: df.describe()
```

```
Out[7]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

Exploration Summary:

1. The dataset contains 9,827 rows and 29 columns, showing good consistency.
2. The dataset is clean, with no missing (NaN) values or duplicate records.
3. The release_date column should be converted to the datetime format, and the year should be extracted for analysis.
4. The overview, original_language, and poster_url columns are not required for this analysis and should be removed.
5. There are outliers in the popularity column, which should be handled before analysis.
6. The vote_average column can be categorized into different rating groups for better analysis.
7. The genres column contains comma-separated values and extra whitespace, which should be cleaned and converted into a categorical format.

```
In [8]: df.head()
```

Out[8]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language
--	--------------	-------	----------	------------	------------	--------------	-------------------

0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en

In [9]:

```
df['Release_Date'] = pd.to_datetime(df['Release_Date'])
print(df['Release_Date'].dtype)
```

datetime64[ns]

In [10]:

```
df['Release_Date'] = df['Release_Date'].dt.year
df['Release_Date'].dtypes
```

Out[10]: dtype('int32')

In [11]:

```
df.head()
```

Out[11]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	

Dropping the Columns

In [12]:

In [15]: `cols = ['Overview', 'Original_language', 'Poster_url']`

In [17]: `print(df.columns.tolist())`

```
['Release_Date', 'Title', 'Overview', 'Popularity', 'Vote_Count', 'Vote_Average', 'Original_Language', 'Genre', 'Poster_Url']
```

In []:

In [24]: `cols = ['overview', 'original_language', 'poster_url']`
`df.drop(columns=cols, inplace=True, errors='ignore')`

In [25]: `print(df.columns.tolist())`

```
['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average', 'Genre']
```

In [26]: `df.head()`

Out[26]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	
0	2021	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure,
1	2022	The Batman	3827.658	1151	8.1	Crime, Mystery
2	2022	No Exit	2618.087	122	6.3	
3	2021	Encanto	2402.201	5076	7.7	Animation, C Family,
4	2021	The King's Man	1895.511	1793	7.0	Action, Adventure,

Categorize the vote_average Column

The vote_average column is categorized into four groups: Not Popular, Below Average, Average, Popular. This categorization makes it easier to analyze and compare movies based on their rating categories are created using the pd.cut() function, which groups the continuous vote_average values into defined rating ranges.

```
In [27]: def categorize_col(df, col, labels):  
  
    edges = [df[col].describe()['min'],  
             df[col].describe()['25%'],  
             df[col].describe()['50%'],  
             df[col].describe()['75%'],  
             df[col].describe()['max']]  
    df[col] = pd.cut(df[col], edges, labels = labels, duplicates = 'drop')  
    return df
```

```
In [28]: labels = ['not_popular', 'below_avg', 'average', 'popular']  
  
    categorize_col(df, 'Vote_Average', labels)  
  
    df['Vote_Average'].unique()
```

```
Out[28]: ['popular', 'below_avg', 'average', 'not_popular', NaN]  
Categories (4, object): ['not_popular' < 'below_avg' < 'average' < 'popular']
```

```
In [29]: df.head()
```

```
Out[29]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure,
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery
2	2022	No Exit	2618.087	122	below_avg	
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family,
4	2021	The King's Man	1895.511	1793	average	Action, Adventure,

```
In [30]: df['Vote_Average'].value_counts()
```

```
Out[30]: Vote_Average  
not_popular    2467  
popular        2450  
average        2412  
below_avg      2398  
Name: count, dtype: int64
```

```
In [31]: df.dropna(inplace = True)  
  
    df.isna().sum()
```

```
Out[31]: Release_Date    0
         Title          0
         Popularity     0
         Vote_Count     0
         Vote_Average   0
         Genre          0
         dtype: int64
```

```
In [32]: df.head()
```

```
Out[32]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure,
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery
2	2022	No Exit	2618.087	122	below_avg	
3	2021	Encanto	2402.201	5076	popular	Animation, (Family,
4	2021	The King's Man	1895.511	1793	average	Action, Adventure,

```
In [33]: df['Genre'] = df['Genre'].str.split(', ')

df = df.explode('Genre').reset_index(drop=True)

df.head()
```

```
Out[33]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Gen
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Actio
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventu
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fictio
3	2022	The Batman	3827.658	1151	popular	Crin
4	2022	The Batman	3827.658	1151	popular	Myste

```
In [52]: # Casting column into Cateogy

df['Genre'] = df['Genre'].astype('category')

df['Genre'].dtypes
```

```
Out[52]: CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
                                     'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                                     'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                                     'TV Movie', 'Thriller', 'War', 'Western'],
                        , ordered=False, categories_dtype=object)
```

```
In [35]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Release_Date 25552 non-null  int32
1   Title        25552 non-null  object
2   Popularity   25552 non-null  float64
3   Vote_Count   25552 non-null  int64
4   Vote_Average 25552 non-null  category
5   Genre        25552 non-null  category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB
```

```
In [36]: df.nunique()
```

```
Out[36]: Release_Date    100
         Title          9415
         Popularity     8088
         Vote_Count     3265
         Vote_Average      4
         Genre          19
         dtype: int64
```

```
In [37]: df.head()
```

```
Out[37]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

Data Visualization

```
In [38]: sns.set_style('whitegrid')
```

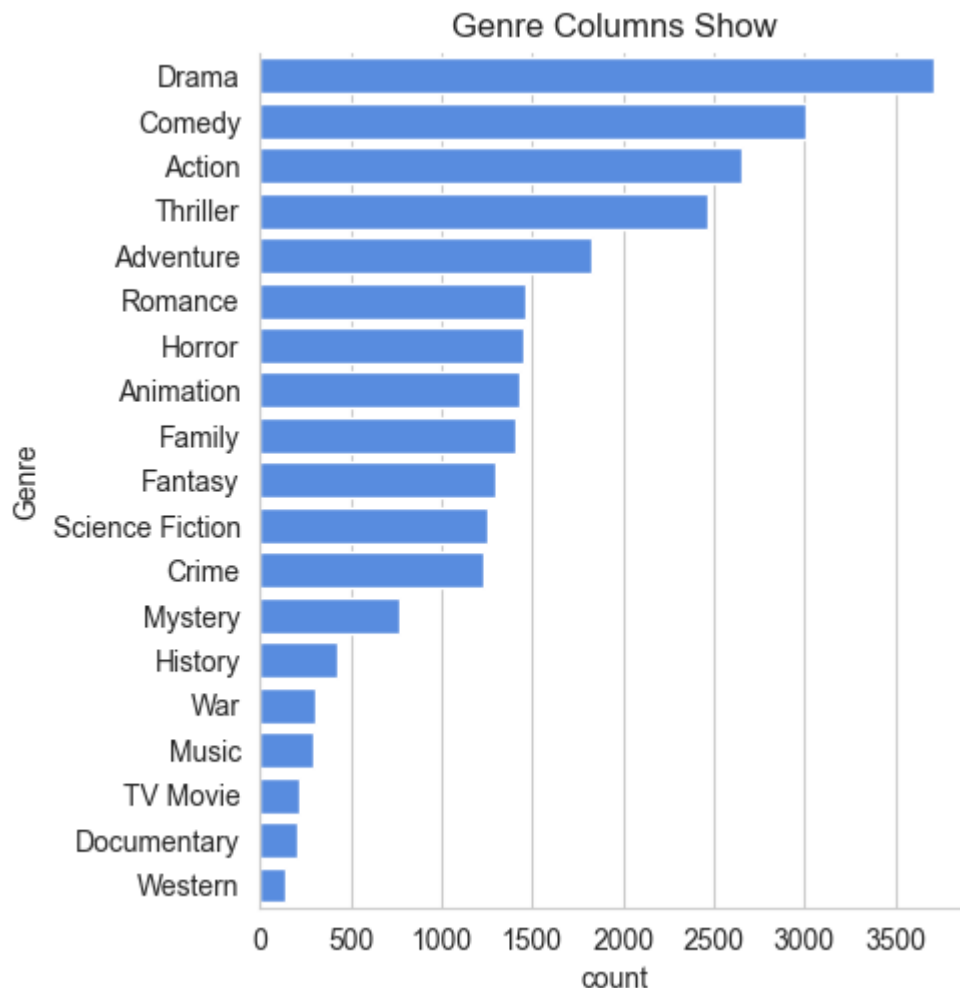
What is the most frequent genre of movies released on Netflix?

```
In [40]: df['Genre'].describe()
```

```
Out[40]: count    25552
         unique      19
         top      Drama
         freq     3715
         Name: Genre, dtype: object
```

```
In [43]: sns.catplot(y = 'Genre', data = df, kind = 'count', order = df['Genre'].value_counts().index)

plt.title("Genre Columns Show")
plt.show()
```



In []:

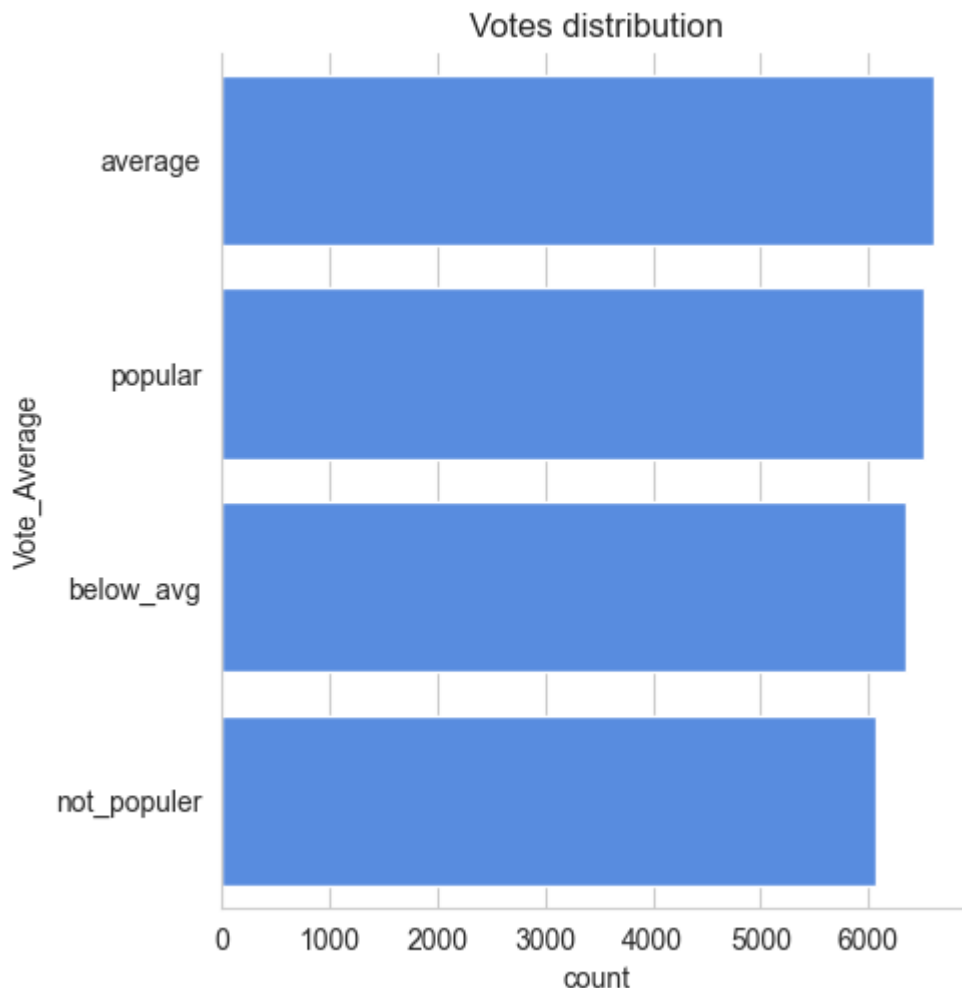
Which has highest votes in vote avg column?

In [44]: `df.head()`

Out[44]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Gen
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Actic
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventu
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fictic
3	2022	The Batman	3827.658	1151	popular	Crin
4	2022	The Batman	3827.658	1151	popular	Myste

In [47]: `sns.catplot(y = 'Vote_Average', data = df, kind = 'count', order = df['Vote_Average']
plt.title("Votes distribution")
plt.show()`



In []:

What movie got the highest popularity? what's genre?

In [49]: `df[df['Popularity'] == df['Popularity'].max()]`

Out[49]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Gen
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Actic
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventu
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fictic

In []:

What movie got the lowest popularity? what's genre?

In [50]: `df[df['Popularity'] == df['Popularity'].min()]`

Out[50]:

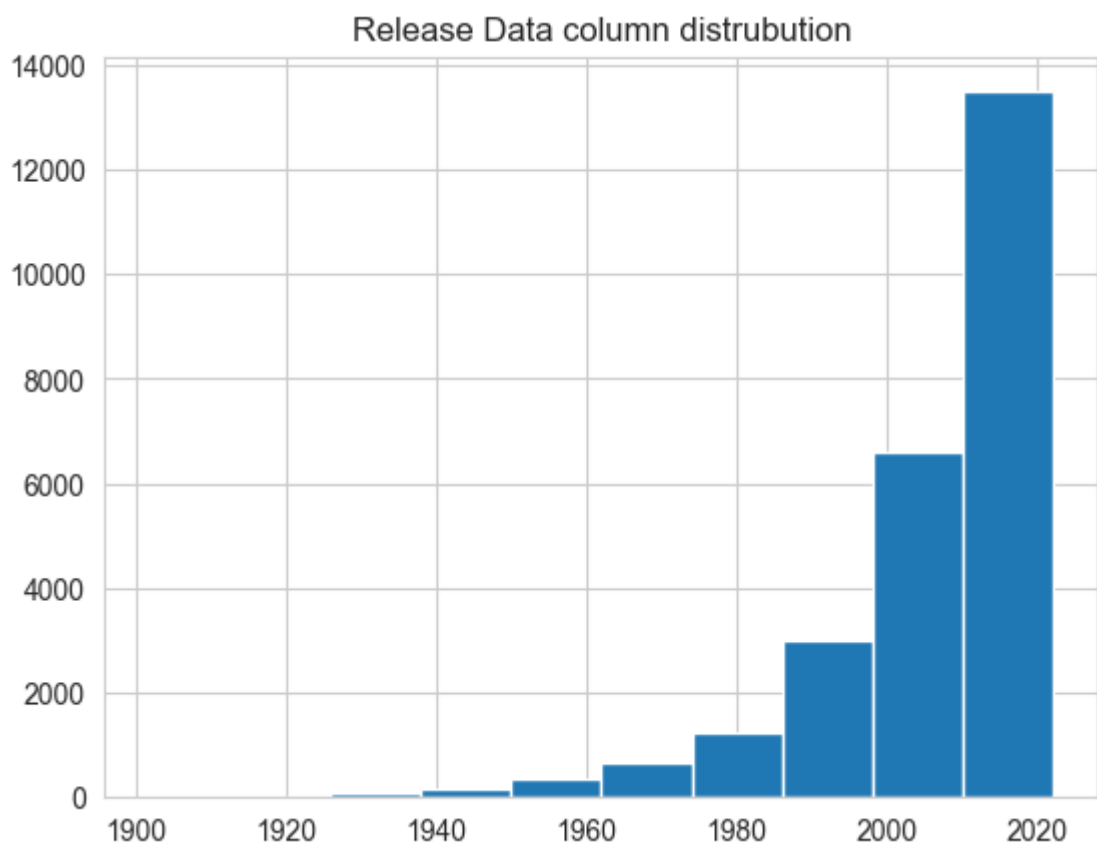
	Release_Date	Title	Popularity	Vote_Count	Vote_Average
25546	2021	The United States vs. Billie Holiday	13.354	152	average
25547	2021	The United States vs. Billie Holiday	13.354	152	average
25548	2021	The United States vs. Billie Holiday	13.354	152	average
25549	1984	Threads	13.354	186	popular
25550	1984	Threads	13.354	186	popular
25551	1984	Threads	13.354	186	popular

In []:

Which year has the most filmed movies?

In [51]:

```
df['Release_Date'].hist()
plt.title('Release Data column distrubution')
plt.show()
```



In []:

Conclusion

1. What is the most frequent genre in the dataset? Answer: Drama is the most frequent genre in the dataset, appearing in more than 40% of the movies.

2. Which genre has the highest votes? Answer: Around 25.5% of the dataset (6,520 movies) fall the Popular vote category. Among these, Drama has the highest number of highly rated movies accounting for more than 18.5%.
3. Which movie has the highest popularity, and what is its genre? Answer: Spider-Man: No Way Home has the highest popularity score in the dataset. Its genres are Action, Adventure, and Science Fiction.
4. Which movie has the lowest popularity, and what is its genre? Answer: The United States vs. John Holmes has the lowest popularity score in the dataset. Its genres include Music, Drama, and Crime.
5. Which year has the highest number of released movies? Answer: 2020 has the highest number of movie releases in the dataset.

In []: